

Lot Summary and Final MRB State

LOT	PRODUCT	FLOW	LOTSIZE	MRB
Q8R7-22	AMX-4100	LVT gate	12 wafers	2026-06-11

Disposition

Release wafers 01, 02, 04, 06, 08, 10, and 11. Wafer 09 is conditional pending REL-22-A.
Wafers 03, 05, and 12 remain on engineering hold. Wafer 07 is scrap and must not be counted in ship yield.

Wafer	Good die	Defect %	Probe yield	Final state	Note
01	18,420	0.83	97.8	release	center-edge normal
02	18,216	0.86	97.1	release	one edge cluster
03	17,604	1.24	93.8	hold	map quadrant B4
04	18,390	0.81	97.6	release	normal
05	17,118	1.42	91.9	hold	scratch arc
06	18,275	0.88	96.9	release	normal
07	16,902	1.58	90.6	scrap	metal flakes
08	18,301	0.79	97.4	release	normal
09	17,880	1.06	95.5	conditional	param drift
10	18,144	0.90	96.8	release	normal
11	18,013	0.95	96.2	release	normal
12	17,422	1.31	92.7	hold	edge bead

Traveler and Process Route

Step	Tool	Recipe	Start	End	Visible state
Implant P3	IMP-17	BORON_LVT_042	05:12	05:58	accepted
Ash	ASH-04	ASH_STD_18	06:10	06:34	accepted
Wet clean	WET-12	SC1_SC2_09	06:48	07:22	accepted
PVD TIN	PVD-03	TIN_GATE_31	08:06	09:44	alarm at 09:18
Anneal	RTP-08	RTP_LVT_22	10:01	10:29	accepted
Metrology	CDSEM-06	GATE_LVT_QC	10:45	11:56	reviewed

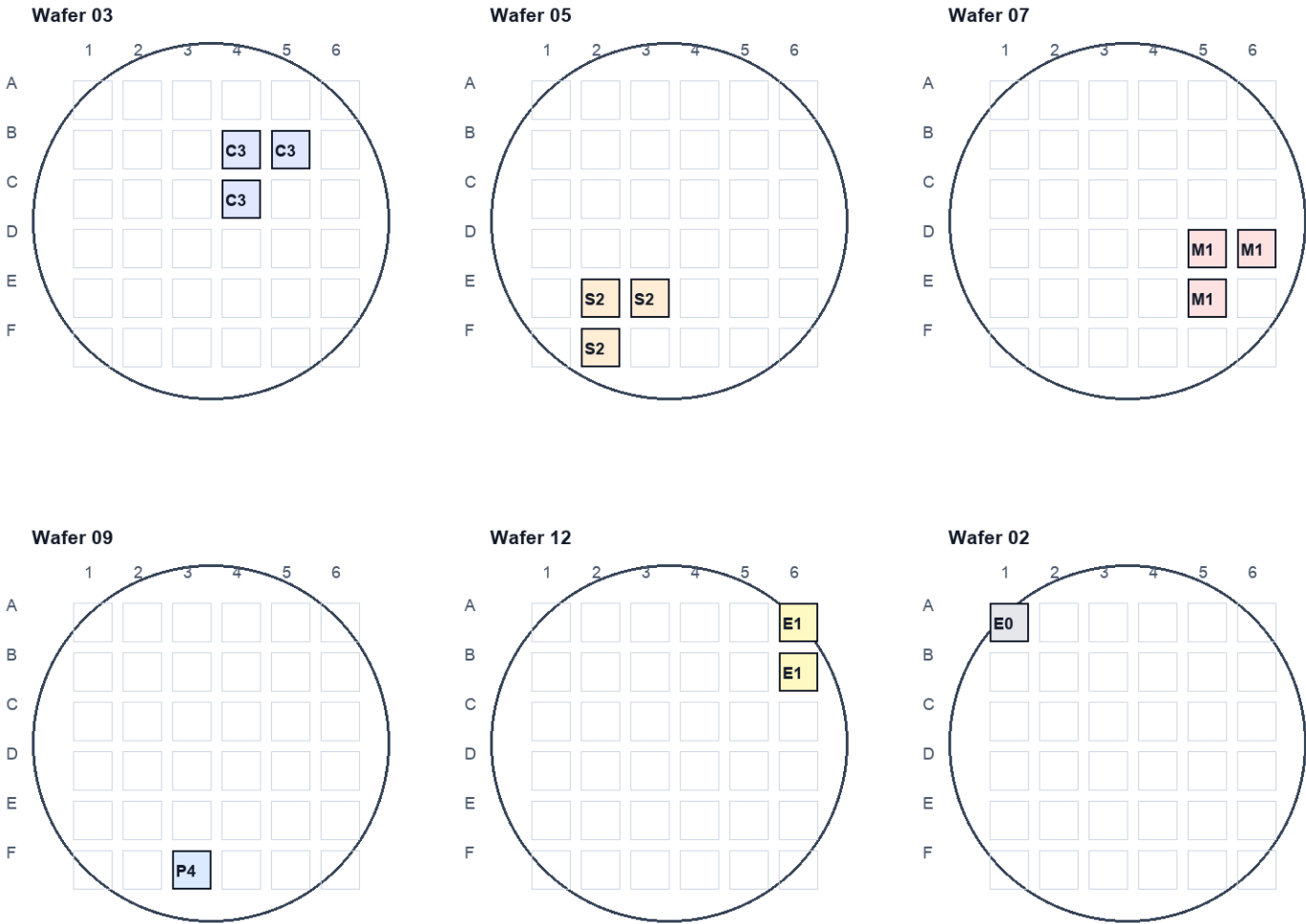
Route note

PVD-03 alarm at 09:18 triggered containment. The lot continued to RTP-08 only after engineering released the tool to complete the already-loaded chamber sequence.

Source	Recipe shown	Status	Meaning
Traveler page	TIN_GATE_31	current	used for lot
Tool alarm printout	TIN_GATE_30	stale header	screen cache
MES audit	TIN_GATE_31	confirmed	controls
Engineer note	increase clamp purge	future only	not used for this lot

Wafer Map Contact Sheet

Each wafer map uses row letters A-F and column numbers 1-6. Filled cells mark reviewed defect clusters; labels show the dominant code at that coordinate.



Code	Visual cue	Critical coordinate	Disposition link
M1	red cells	W07 D5/D6	scrap
S2	orange scratch arc	W05 E2-F2	engineering hold
E1	yellow edge	W12 A6/B6	engineering hold
P4	blue drift	W09 F3	conditional release
C3	violet cluster	W03 B4/C4	engineering hold

CD-SEM Metrology and Flags

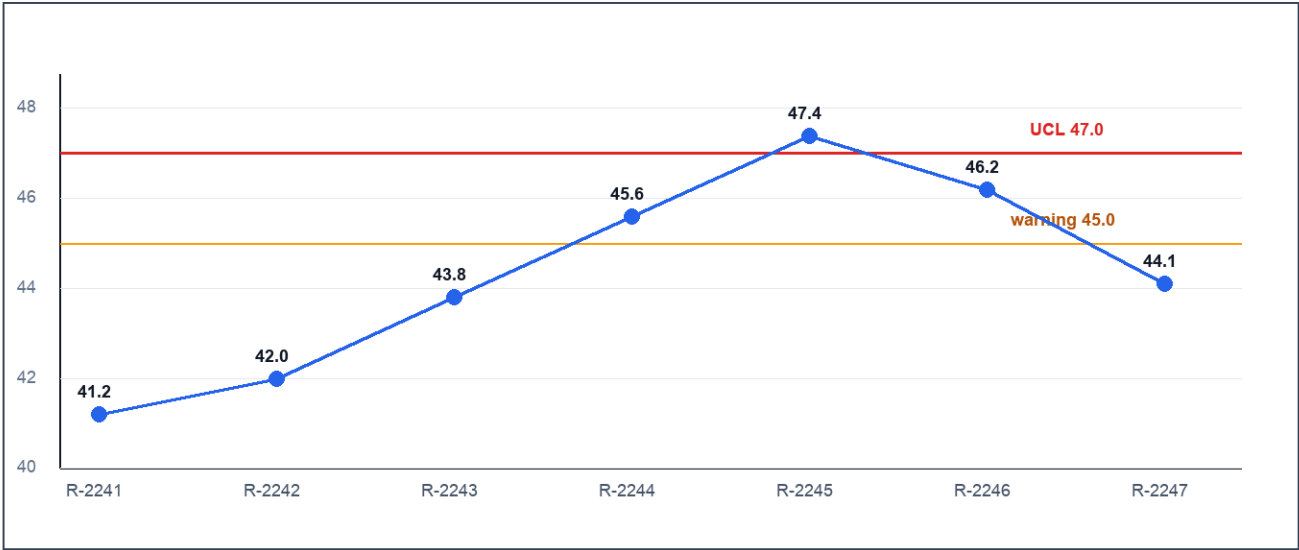
Wafer	Site	CD nm	Delta	Flag	Reviewer note
03	B4	27.8	+2.9	H	matches map cluster
03	C4	27.4	+2.5	H	adjacent high
05	E2	24.1	-1.8	L	scratch path
07	D5	29.6	+4.7	H	flake shadow
07	D6	29.1	+4.2	H	flake shadow
09	F3	26.8	+1.9	watch	param drift only
12	A6	23.9	-2.0	L	edge bead

Specification note

Target gate CD is 24.9 nm with action limits -2.0 nm and +2.5 nm. Sites at or beyond an action limit require MRB disposition even when the wafer-average yield remains above 93%.

Site group	Rows affected	Rule	Final state
High CD cluster	W03 B4/C4 and W07 D5/D6	beyond +2.5 nm	hold or scrap
Low CD edge	W05 E2 and W12 A6	at -1.8/-2.0 nm	hold
Watch drift	W09 F3	+1.9 nm below action	conditional

SPC Trend and PVD Alarm



Run	Time	TiN thickness	Limit state	Comment
R-2241	08:06	41.2	inside	start
R-2242	08:22	42.0	inside	stable
R-2243	08:38	43.8	inside	rising
R-2244	08:54	45.6	warning	near UCL
R-2245	09:10	47.4	above UCL	alarm follows
R-2246	09:26	46.2	warning	post alarm
R-2247	09:42	44.1	inside	recovered

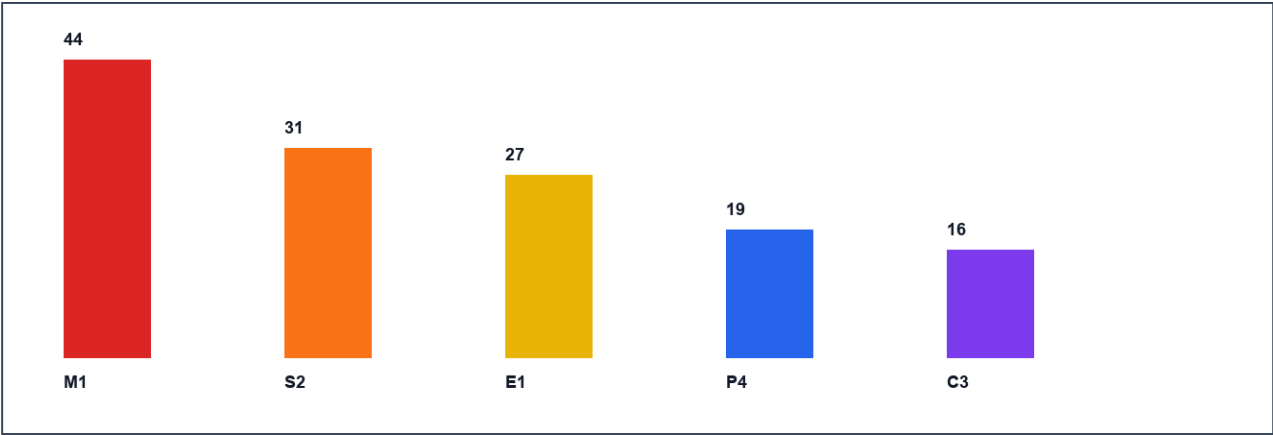
Alarm interpretation

R-2245 is above UCL at 47.4 nm and is the only plotted point beyond the red line. R-2244 and R-2246 are warning-state points, not rejects by themselves.

Interlock	Observed value	Threshold	State	Disposition
PVD-03 chamber pressure	4.8 mTorr	<=5.0 mTorr	pass	not root cause
Clamp purge flow	18.1 slm	>=19.0 slm	fail	future purge increase only
Ti target age	81.4 kWh	<=85 kWh	watch	continue after clean
Thickness monitor residual	+1.7 nm	+/-1.0 nm	fail	drives R-2245 review
Alarm acknowledgment	09:18 by J. Kwon	required	complete	containment clock starts

Defect Pareto and Classification

Code	Class	Count	Affected wafer	Disposition
M1	metal flake	44	07	scrap wafer 07
S2	scratch	31	05	hold for review
E1	edge bead	27	12	hold
P4	parametric drift	19	09	conditional release
C3	center cluster	16	03	hold



Gate	Trigger	Wafer/site	Required action	Owner
DCT-1	M1 count > 20	W07 D5/D6/E5	scrap wafer; do not sample for REL	MRB
DCT-2	S2 scratch crossing active cells	W05 E2/F2/E3	engineering hold; inspect sister lot Q8R7-23	Yield Eng
DCT-3	E1 edge bead two adjacent cells	W12 A6/B6	hold pending edge-bead review	Process Eng
DCT-4	P4 drift with clean SEM image	W09 F3	conditional release; REL-22-A required	Reliability
DCT-5	C3 high-CD cluster	W03 B4/C4/B5	hold pending CD remeasure	Metrology

Classification rule

M1 metal flake is a scrap class when clustered on adjacent D-row sites. S2, E1, and C3 require engineering hold. P4 is allowed only as conditional release with reliability sample.

Recipe Source Conflict

Source	Recipe shown	Status	Meaning
Traveler page	TIN_GATE_31	current	used for lot
Tool alarm printout	TIN_GATE_30	stale header	screen cache
MES audit	TIN_GATE_31	confirmed	controls
Engineer note	increase clamp purge	future only	not used for this lot

Local note

The tool printout shows TIN_GATE_30 only because its header cache was stale. MES audit confirms TIN_GATE_31 controlled the actual run.

MRB Disposition Matrix

Item	Decision	Reason	Owner
Wafer 07	scrap	metal flakes M1 at D5/D6	Y. Nishida
Wafer 05	hold	scratch S2 crosses E2-F2	A. Roy
Wafer 03	hold	CD high at B4/C4	M. Alvarez
Wafer 12	hold	edge bead at A6	K. Stone
Wafer 09	conditional release	parametric drift P4, REL sample required	S. Han
Wafers 01,02,04,06,08,10,11	release	within disposition limits	MRB

Local note

The MRB matrix is the controlling disposition source for wafer-level state.

Reliability Sample Plan

Sample	Wafer	Stress	Result
REL-22-A	09	HTOL 168h	pending
REL-22-B	02	TC 100 cyc	pass
REL-22-C	11	HAST 96h	pass
REL-22-D	07	not built	scrap

Local note

Wafer 09 conditional release depends on REL-22-A HTOL 168h passing; wafer 07 gets no reliability credit because it is scrap.

Shipping Allocation Worksheet

Ship bucket	Wafers	Die count	Condition
Release now	01,02,04,06,08,10,11	127,759	standard COA
Conditional	09	17,880	hold shipment until REL-22-A pass
Engineering hold	03,05,12	52,144	do not ship
Scrap	07	16,902	do not include in yield

Local note

Release-now die count excludes wafer 09 until REL-22-A passes and excludes scrap wafer 07 completely.

Inline Photo Review Register

Frame	Wafer/site	Image cue	Finding	Disposition link
IMG-221	W07 D5	bright flake at lower electrode	M1 metal flake	scrap W07
IMG-222	W07 D6	second adjacent flake	M1 cluster	scrap W07
IMG-223	W05 E2	diagonal drag mark	S2 scratch	hold W05
IMG-224	W12 A6	edge residue	E1 edge bead	hold W12
IMG-225	W09 F3	clean image, param high	P4 electrical drift	conditional W09

Frame	Focus	Measured cue	Disposition impact	Reviewer mark
IMG-221	W07 D5	flake 8.2 um	scrap support	red circle / counted
IMG-222	W07 D6	flake 6.9 um	adjacent M1 cluster	red circle / counted
IMG-223	W05 E2	scratch 118 um	hold support	orange line / counted
IMG-224	W12 A6	edge residue band	hold support	yellow bracket
IMG-225	W09 F3	no visible particle	electrical drift only	blue square / reference

Image archive note

The SEM image files remain in the fab image archive; this signed packet carries the reviewer marks, measured cues, and disposition links used by MRB. The visual cue column must stay attached to the frame and wafer site.

Final Certificate of Analysis Holdback

COA field	Visible value	Applies to
Lot	Q8R7-22	all wafers
Released wafers	01,02,04,06,08,10,11	ship now
Conditional wafer	09	ship only after REL-22-A pass
Engineering hold	03,05,12	excluded from shipment
Scrap	07	excluded from yield and COA
Recipe	TIN_GATE_31	MES-confirmed
Yield denominator	released + conditional only	do not include W07

Certificate note

The preliminary COA draft listed wafers 01-12, but the signed COA must list only released wafers and the conditional wafer separately. The draft is not the shipping authority.

Disposition Recap and Audit Trail

Timestamp	Actor	Entry	Effect
2026-06-10 12:18	MES	PVD alarm attached	containment starts
2026-06-10 14:40	Yield eng	wafer map review complete	W03/W05/W07/W12 flagged
2026-06-11 08:55	MRB	W07 scrap, W09 conditional	controls final state
2026-06-11 10:20	Quality	COA holdback added	prevents draft all-wafer ship
2026-06-11 13:30	Planning	ship bucket split	127,759 die release now

Control item	Final state	Reason
Ship now	wafers 01,02,04,06,08,10,11	released by MRB
Conditional	wafer 09	REL-22-A pending
Hold	wafers 03,05,12	CD/scratch/edge issues
Scrap	wafer 07	M1 metal flake cluster
Recipe conflict	TIN_GATE_31 controls	MES audit beats stale printout